NSDDD Search Interface Workshop

Searching 671 Documents from 118 Countries with Semantic Embeddings

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Edinburgh National Security and Defence Documents Dataset v3.5 (2026)

What Is the NSDDD?

671
Documents

118
Countries

1987-2025
Time span

787,844
Text segments

Document types: 290 Defence Documents | 204 National Security Strategies | 170 Defence White Papers | 7 Threat Assessments
Metadata: 26 international organisations, UN regions, Freedom House, World Bank income, ODA status, Huntington civilisations, small states classifications

Neal, A. W. and Gardner, R. B. (2026). NSDDD v3.5. University of Edinburgh. doi.org/10.7488/ds/8101

What Is an Embedding Model?

Core concept

A model that converts text into a fixed-length numerical representation: a vector. Every sentence in the corpus has been passed through all-mpnet-base-v2 and converted into a list of 768 numbers that encode its meaning.

'Climate change poses a threat to national security' → **all-mpnet base-v2** → `[0.023, -0.187, 0.412, ..., 0.089, -0.034, 0.156]`

Input sentence | Embedding model | 768-dimensional vector

What Are Vectors?

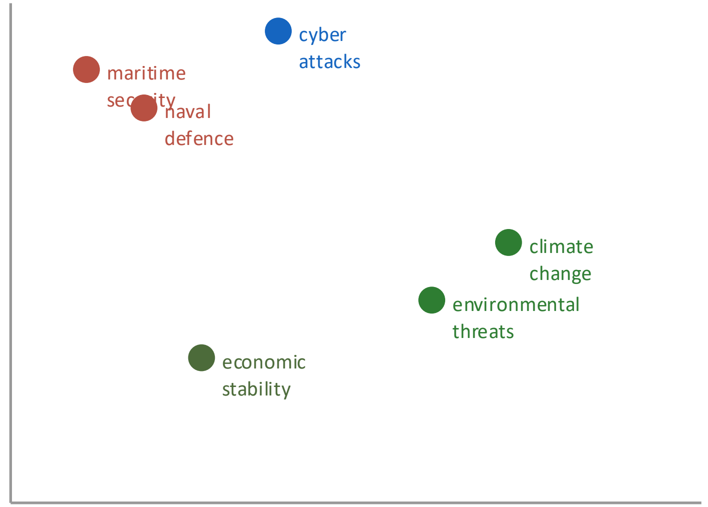
A GPS coordinate for meaning

Just as latitude and longitude locate a place on earth, a 768-number vector locates a sentence in meaning-space.

In two dimensions, you can plot sentences on an X/Y grid. ‘Maritime security’ and ‘naval defence’ land close together because they mean similar things. ‘Climate change’ lands somewhere else.

The model does this in 768 dimensions, which is impossible to visualise directly, but the principle is the same.

Simplified 2D meaning-space



Sentences close together in this space are similar in meaning

It Is an LLM, but Not Generative AI

What it is

- 110 million parameters
- Trained on massive text corpora
- Understands language deeply
- Takes a sentence in, produces a vector out
- **A measuring instrument**

What it is not

- Does not produce text
- Does not predict the next word
- Does not hallucinate
- Is not an author or analyst
- **Results are real sentences from real documents**

The search results are always drawn from real sentences in real government documents. Nothing is invented.

Semantic Search and Why Full Sentences Matter

The search pipeline

1. You type a query as a full sentence
2. The query is encoded into a 768-dim vector
3. Compared against all 787,844 pre-encoded segments
4. Cosine similarity produces a score (0–1)
5. Results above threshold returned, ranked by score
6. Results clustered and named using TF-IDF

Why full sentences?

The model was trained on sentence pairs. Single words produce imprecise vectors.

‘cyber’ → vague; matches everything digital

‘cyber attacks on critical national infrastructure’ → precise; finds hacking, infrastructure vulnerability, digital resilience

The sentence you type is a description of the meaning you are looking for.

For a specific word or phrase, use keyword search instead.

Think carefully about what meaning you are searching for. Different phrasings find different things.

This Is Not a Chatbot: Do Not Ask it Questions

What it cannot do

The system does not understand questions. It does not interpret intent. It does not produce answers.

Typing a question into the search box does not produce an answer to that question. It produces text segments from government documents that are semantically close to the wording of your question.

This is a fundamentally different operation.

Questions produce poor results because interrogative constructions are rare in national security documents. The model cannot find strong matches for sentence structures that barely appear in the corpus.

What to do instead

Rephrase your question as a declarative statement describing the meaning you want to find.

✗ "What do states say about climate change?"

✓ "Climate change poses a threat to national security"

✗ "How do governments approach cyber security?"

✓ "Cyber attacks threaten critical national infrastructure"

The query is not a question to be answered. It is a description of the meaning you are searching for.

Do not type questions. Type the meaning you are looking for as a full declarative sentence.

Query Construction: Avoid Compound Sentences

The problem

A compound query mixes two distinct meanings into a single vector. The model averages the semantic content, and the resulting vector points at neither idea with precision.

✗ "Climate change is a threat and cyber attacks are a growing risk"

This hovers between climate security and cyber security. Results will be a mixed set that satisfies neither query well.

✗ "Terrorism and organised crime undermine state stability and erode civil society"

Four concepts in one sentence. The results will be incoherent.

The rule and good examples

One concept per query. Run separate searches and compare results.

Good query construction:

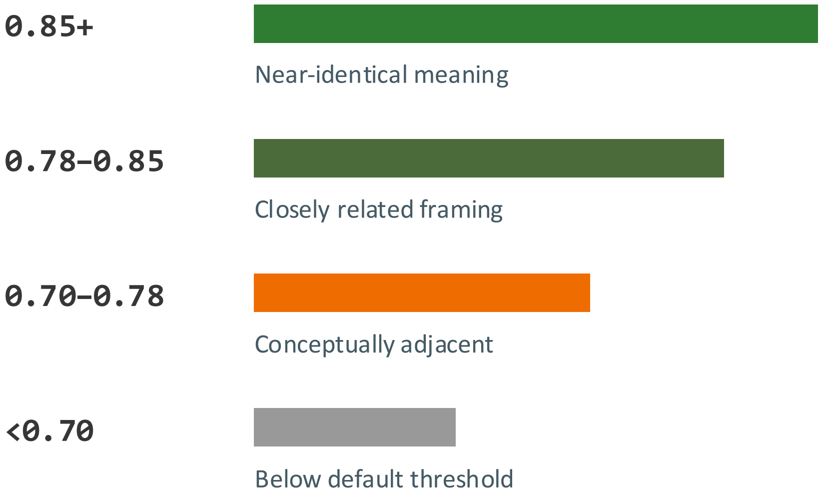
- ✓ "Climate change is a threat to national security"
- ✓ "Cyber attacks on critical national infrastructure"
- ✓ "Terrorism destabilises state authority"
- ✓ "Organised crime undermines civil society"

Each query is a single clear declarative sentence describing one concept. The resulting vector is precise.

One concept, one query. If your sentence contains 'and' joining two ideas, split it into two searches.

The Similarity Score

What the score means



For quantitative work

The score is a continuous variable, not just a binary in/out filter.

The threshold you set shapes the scope of your analysis.

0.70 = broad conceptual net
0.85 = strict, precise matches only

As you move down the results list, the meaning shifts gradually. This gradient is itself informative.

How Far Down the Results List to Go

The distribution of results

Results are ranked by semantic similarity from highest to lowest. The most relevant segments appear first, but relevant material can appear much further down the list, buried among lower-scoring segments.

Semantic search does not produce a clean boundary between relevant and irrelevant. A result at rank 80 with a score of 0.71 may be directly relevant to your research question. A result at rank 5 with a score of 0.83 may be technically close but not useful for your specific inquiry.

The similarity score is a measure of semantic distance from your query, not a quality judgement.

Approaches

There is no single correct cutoff. Consider these approaches:

Score-based: review all results above a chosen threshold (e.g. 0.75). Systematic, but may still include noise. Different topics may need different thresholds, depending on results.

Proportion-based: scroll until the majority of consecutive results are no longer relevant. Stop when more than half are off-topic.

Manual review: export the CSV, read through all results, and mark each as accept or reject.

For quantitative work, document and justify your cutoff threshold so the analysis is reproducible.

Relevant results can sit deep in the list. Review systematically. Document your cutoff threshold for reproducibility.

Cautions for Quantitative Work ⚠️

Unequal document counts

Norway has 27 documents. Some countries have one. Cross-country comparisons must account for this.

Unequal document length

Japan publishes exceptionally long annual defence documents. The system works at the sentence level, so longer documents contribute more sentences.

Unequal time coverage

Some countries have documents spanning decades; others entered the dataset recently. Time-series analysis needs care.

Sentence counts ≠ document prevalence

Raw counts can be pulled up by repeated mentions within a single document. Use document-level prevalence, not sentence counts, for trend analysis.

The Visualisation Outputs

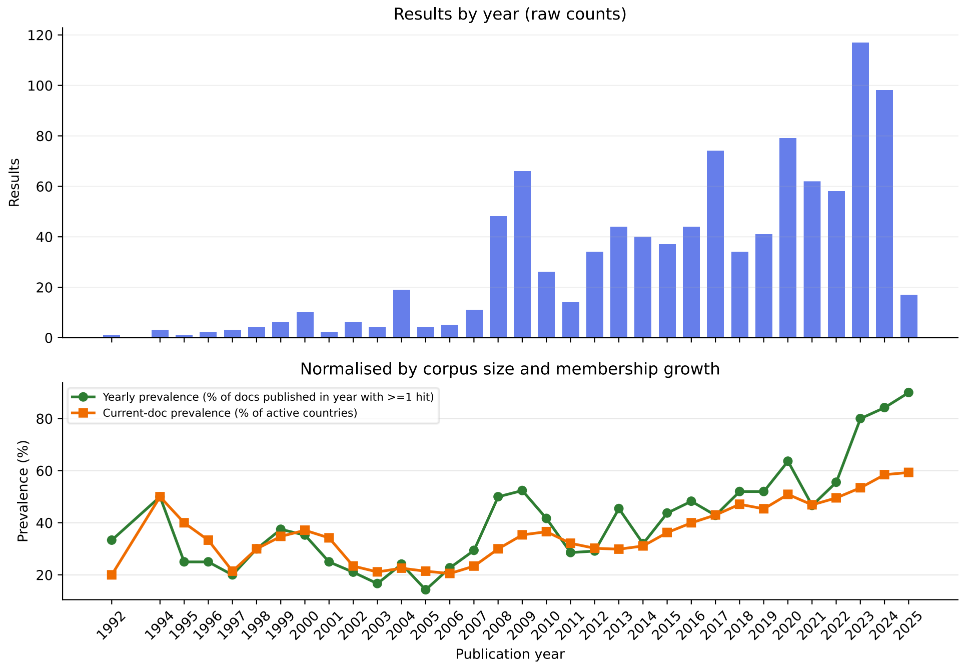
Year chart (two panels)

Top panel: Raw segment counts per year. Can be misleading without context.

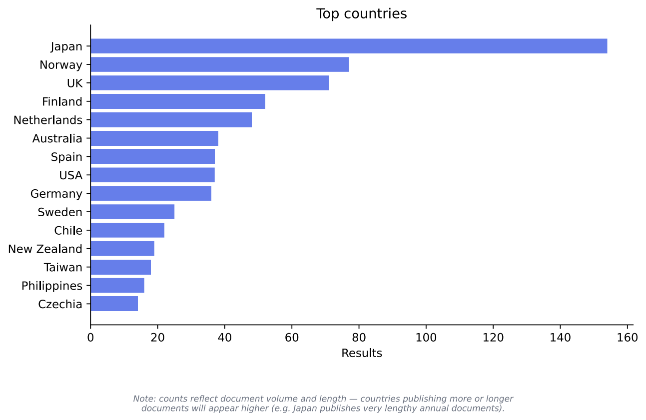
Green line: Yearly prevalence. Share of documents published in that year with at least one hit.

Orange line: Current-document prevalence. Share of active countries whose latest document mentions the topic.

These two lines tell different stories: one about publishing trends, one about diffusion across the active corpus.

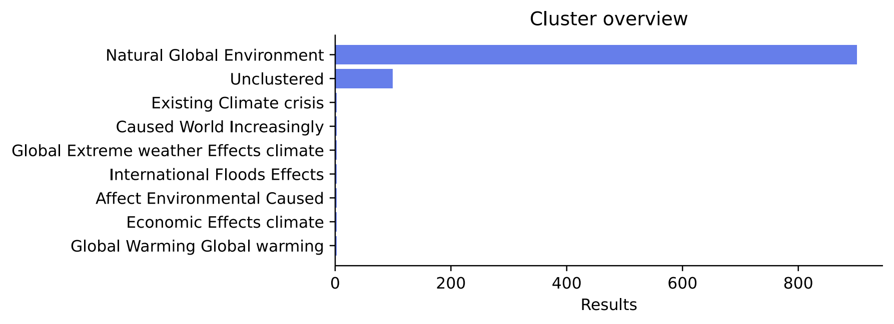


The Visualisation Outputs



Country chart

Top 15 countries by raw count. Carries an explicit caveat about document volume and length.



Cluster chart

Shows how results group by semantic similarity. Cluster names are auto-generated using TF-IDF, excluding search terms to highlight what distinguishes each group.

A Note on Clustering

What clustering does

The clustering pipeline groups returned segments by semantic similarity and assigns auto-generated labels using TF-IDF. This helps you see sub-themes in a large result set without reading every segment individually.

Cluster names are generated from the most distinctive words in each group, with your search terms excluded so the labels highlight what distinguishes the clusters from each other.

Clustering is off by default in Colab for speed. Enable it in settings if you want cluster labels and the cluster chart.

Caveats

Clustering does not affect the underlying similarity scores or the CSV export. It is an exploratory navigation aid, not an analytical method.

Cluster assignments are merely heuristic labels.

Do not treat cluster membership as a classification. It is a starting point for closer reading.

For any quantitative use of clusters, validate manually. Review the segments in each cluster and confirm they are coherent before using cluster categories in analysis.

Clustering helps navigate large result sets. It is off by default for speed. Do not treat cluster assignments as analytical conclusions.

Qualitative Work with the Exported Data

The CSV export

Columns: cluster ID, cluster name, score, segment text, context before/after, document name, year, country, ISO code, region, subregion, income group, freedom status, plus ‘Accept result’ and ‘Notes’ for your own coding.

The whole-of-government example

A search for ‘whole-of-government approach to security’ found the full conceptual genealogy of this idea:

‘Broad, holistic approach’ (South Africa 1998) → ‘comprehensive approach’ (NZ 2000) → ‘integrated/coordinated’ (Canada, Singapore, Uganda, Finland 2004) → ‘whole-of-government’ crystallises (Australia 2005) → ‘whole-of-society’, ‘whole-of-nation’ (recent)

By 2025, the term spans 55 documents and 30 countries. The semantic search found all of these variants because they are close in meaning-space, even though they use different words. A keyword search would have missed most of them.

Manually colour-coding the results in the exported spreadsheet revealed the range of formulations over geopolitical space and time.

Qualitative Work with the Exported Data

	A	B	C	D
1	Topic	whole of government approach to security		
2	Search type	Semantic		
3	Search threshold	0.7		
4	Cluster threshold	0.78		
5	Number of search results	2560		
6	Number of clusters	27		
7	Context window	1		
8	Active filters	Years: 1987–2025		
9				
10	Cluster ID	Cluster Name	Search score	Segment text
11	cluster_0	Policy Strategy Security policy	0.8456	Guaranteeing a Whole-of-Government Approach to Security
12	cluster_0	Policy Strategy Security policy	0.8229	These circumstances necessitate an effective whole-of-government approach to security
13	cluster_0	Policy Strategy Security policy	0.8181	Whole of Government Responses Current threats to security require a whole-of-government approach
14	cluster_0	Policy Strategy Security policy	0.8097	A whole-of-government approach to security – dealing with emergencies, disasters and new threats is a task we all share
15	cluster_0	Policy Strategy Security policy	0.8094	A whole-of-Government approach towards national security is needed to reduce duplication of effort and to increase inter-sectoral coordination
16	cluster_0	Policy Strategy Security policy	0.8071	What is the relationship between Defence and other government agencies to enhance a 'whole of Government' approach towards achieving secur
17	cluster_0	Policy Strategy Security policy	0.8052	It sets out a 'whole of government' approach, based on a concept of security that goes beyond military effects
18	cluster_0	Policy Strategy Security policy	0.8027	The watchword of this Security Strategy is a whole-of-government and whole-of-society approach
19	cluster_0	Policy Strategy Security policy	0.796	This is why a resolute approach to ensuring security must be conceived and carried out in a whole-of-government manner
20	cluster_0	Policy Strategy Security policy	0.7936	It is important to note that the policy provides a whole-of-government approach to address and strengthen national security
21	cluster_0	Policy Strategy Security policy	0.7933	The authors applied the whole-of-government approach to the text, which has improved our understanding of the breadth and complexity of secur
22	cluster_0	Policy Strategy Security policy	0.7926	As a necessary foundation for an effective defence policy, the importance of whole- of-government approaches to the security of the State in its int
23	cluster_0	Policy Strategy Security policy	0.7923	A whole-of-government approach is needed to reduce and eliminate threats to national security
24	cluster_0	Policy Strategy Security policy	0.7913	The Government will work to promote a comprehensive approach to security
25	cluster_0	Policy Strategy Security policy	0.7904	As a whole of government approach, this review provides a common perspective for all elements of government involved in the security of the State
26	cluster_0	Policy Strategy Security policy	0.7899	This whole-of-government framework includes a wide range of government players in understanding the threats, planning responses and impleme
27	cluster_0	Policy Strategy Security policy	0.7844	We need a whole-of-government approach to implementing this National Security Strategy
28	cluster_0	Policy Strategy Security policy	0.7843	The comprehensive interpretation of the concept of security makes it indispensable for its political, military, economic, financial, environmental a
29	cluster_0	Policy Strategy Security policy	0.7838	A MODERNISED TOTAL DEFENCE CONCEPT Our overall defence effort relies heavily on a whole-of-government approach
30	cluster_0	Policy Strategy Security policy	0.7837	e. Internal security Internal security and civil protection require a whole-of-government approach

Metadata Filtering: In-Tool and in Excel

In-tool filters (before searching)

- Region, subregion, individual countries
- 26 international organisations
- Income group, democracy, ODA status
- Small states, S. Huntington's 'civilizations'
- Document type, year range
- Most recent document per country only

Spreadsheet filters (after searching)

The CSV includes columns for country, ISO code, region, subregion, income group, freedom status.

You can sort and filter on any of these in a standard Excel table after the search.

For many use cases, this is simpler than pre-filtering.

When to use which

Use in-tool filters when you want to restrict the search scope (e.g. only search NATO members, or only post-2010 documents). This affects both results and performance.

Use spreadsheet filters when you want to explore and slice the full results after the fact.

Both are valid and complementary.

The document_metadata CSV

What is in the file

One row per document in the corpus. Columns include: document filename, country, ISO code, year, document type, language, word count, and the full set of metadata variables (UN region, subregion, income group, Freedom House score, ODA status, Huntington civilisation, small state classifications, and 26 international organisation memberships).

This file is the reference layer for the corpus. Use it to check coverage, identify gaps, or build custom filters before searching.

Where to find it

The file is included in the NSDDD installer repository and is also available as a standalone download from the dataset page on Edinburgh DataShare.

It is the definitive record of what is in the corpus and should be consulted before designing any comparative or cross-national analysis.

Check the metadata file before designing cross-national comparisons. Coverage is uneven; the file shows you exactly what you have.

Normalisation Context

Sort at will

You need to decide which metrics are meaningful for your analysis.

Topic: ‘climate change is a threat’

Japan has the most hits but also has the longest and most documents in these results.

Poland was the first to discuss this topic (1992), while France has discussed it for longest (1994-2025).

The Cook Islands (and other low-lying island countries) have the highest hits/1000 words: a potential measure of salience.

Climate_change_is_a_threat_norm_context_20260414_131124.csv

ⓘ Normalisation context — matched documents in these results

Sorted by hits. All data are computed from matched documents cited in these results only. Years covered and Avg doc length reflect matched documents only. Coverage% = matched docs ÷ total docs in corpus (full corpus for that country). Hits/1kw = hits ÷ (avg matched doc length × total docs in corpus ÷ 1000). Mean score is the average cosine similarity of matched segments. Click any column heading to sort.

Country	Hits	Matched docs	Coverage %	Mean score	Hits / 1kw	Docs in dataset	Years covered ▲	Avg doc length (words)
Poland	8	4	36%	0.729	0.02	11	1992–2020	36,200
USA	37	12	39%	0.725	0.05	31	1994–2022	24,669
Germany	36	6	75%	0.728	0.20	8	1994–2023	22,877
France	9	4	57%	0.714	0.02	7	1994–2025	55,154
UK	71	12	67%	0.730	0.10	18	1998–2025	37,822
Mongolia	1	1	25%	0.708	0.01	4	1998–1998	28,763
Norway	77	16	59%	0.732	0.04	27	1999–2025	65,381
Czechia	14	6	30%	0.722	0.07	20	1999–2023	9,981
Republic of Korea	11	8	62%	0.722	0.01	13	2000–2023	106,056
Spain	37	7	70%	0.730	0.16	10	2000–2021	23,862
Ireland	11	4	80%	0.730	0.06	5	2000–2019	36,820
Cambodia	8	3	75%	0.723	0.13	4	2000–2022	15,743
Turkey	1	1	100%	0.728	0.02	1	2000–2000	44,279
Finland	52	11	73%	0.721	0.10	15	2001–2025	34,238
Taiwan	18	10	40%	0.731	0.01	25	2002–2023	61,612
Lithuania	8	5	29%	0.709	0.03	17	2002–2023	14,842
Ecuador	7	4	67%	0.717	0.04	6	2002–2019	27,876
Italy	2	2	40%	0.708	0.01	5	2002–2022	77,283
Indonesia	6	4	100%	0.725	0.06	4	2003–2021	25,231
Portugal	1	1	25%	0.708	0.06	4	2003–2003	4,333
Slovenia	10	5	62%	0.734	0.09	8	2004–2020	14,642
Hungary	5	4	50%	0.731	0.06	8	2004–2020	11,132

Three Search Modes

Semantic search (default)

Finds meaning, not words. Best with full sentences. Uses the embedding model to compare your query against the corpus.

Keyword search

Regex-based exact matching. Supports wildcards (* and ?) and fuzzy matching. Use when you want a specific word or phrase.

Boolean search

AND, OR, NOT operators for complex keyword queries. E.g. (cyber OR digital) AND threat AND NOT terrorism.

All three modes cluster results using the same embedding-based clustering pipeline.

Using AI to Analyse Exported Data

Why this is reliable

Ideally, an AI will write code that calls established Python packages (pandas, matplotlib, scipy).

The calculations are executed by those packages, not by the AI. The numbers are correct.

You can inspect the code, rerun it, and verify every step.

Why this is risky

The AI does not know the corpus. It does not know about unequal document counts, lengths, or time coverage.

It will produce charts that are arithmetically correct and analytically misleading.

The discipline required

You need to specify the analysis, not just the question.
Tell the AI to normalise by document count. Tell it to use document-level prevalence. Tell it which countries to control for.

The AI will execute these instructions faithfully, but it may not always supply them on its own.

Use AI to write the code. Design the analysis yourself. Discuss your needs and cautions with the AI.

Try It Yourself

What to search

What concept are you investigating in your own research? Type it as a full sentence.

Topic areas that work well:

- Threat framings: terrorism, cyber, climate, pandemics, hybrid warfare, energy, food, migration
- Institutions: civil-military relations, whole-of-government, oversight, interagency
- Strategic concepts: deterrence, resilience, alliances, sovereignty
- Regional: Arctic, Indo-Pacific, Euro-Atlantic, African peace architecture

Immediate things to try

- Watch the similarity scores shift as you move down the results
- Read the auto-generated cluster names for sub-themes you had not considered
- Filter by region or organisation and compare
- Sort by different columns in the norm_context data
- Open the CSV in Excel and colour-code results by hand to trace terminological variation
- Try the same concept as both a sentence query and a keyword

The goal is not a finished analysis. It is to develop an intuition for how meaning-based search differs from keyword search.

Finding Your Output Files

Outputs are saved to the outputs/ folder inside the NSDDD installation directory.

Default path:
nsddd_v3.5 / outputs /

Each search run outs two CSVs and three SVG visualisations to the outputs folder

Check nsddd_v3.5/outputs for timestamped outputs.

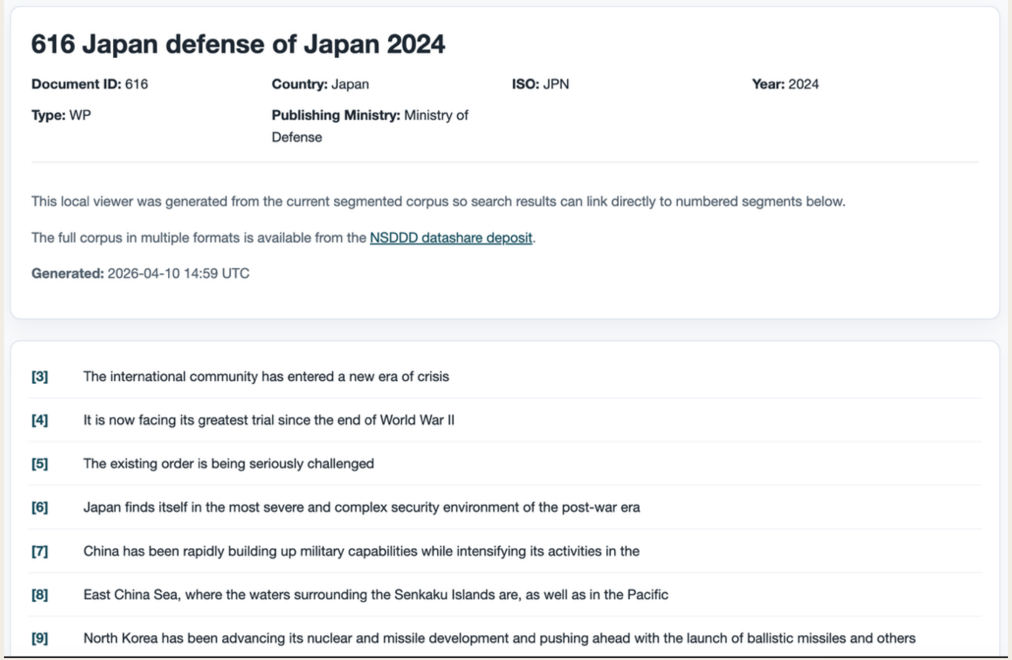
Direct Links to Source Segments

What is new in v3.5

The sample results display and the CSV export now include a direct hyperlink for each returned segment. Clicking the link opens the segment in its full document context within the NSDDD document viewer. This is especially useful for close qualitative analysis.

The CSV column is labelled 'link'. It works as a clickable hyperlink in Excel, Numbers, and Google Sheets.

Documents are in the XHR folder, ordered by prefix number.



616 Japan defense of Japan 2024

Document ID: 616 **Country:** Japan **ISO:** JPN **Year:** 2024

Type: WP **Publishing Ministry:** Ministry of Defense

This local viewer was generated from the current segmented corpus so search results can link directly to numbered segments below.

The full corpus in multiple formats is available from the NSDDD datashare deposit.

Generated: 2026-04-10 14:59 UTC

[3]	The international community has entered a new era of crisis
[4]	It is now facing its greatest trial since the end of World War II
[5]	The existing order is being seriously challenged
[6]	Japan finds itself in the most severe and complex security environment of the post-war era
[7]	China has been rapidly building up military capabilities while intensifying its activities in the
[8]	East China Sea, where the waters surrounding the Senkaku Islands are, as well as in the Pacific
[9]	North Korea has been advancing its nuclear and missile development and pushing ahead with the launch of ballistic missiles and others

Each result links directly to the segment in its original document. Use these links. Context changes meaning.

Installing a CLI LLM to Analyse Results

What is a CLI LLM?

A command-line AI assistant you run in a terminal alongside your notebook. Feed it your exported CSV and ask it to write analysis code, summarise results, draft interpretations or produce visualisations. It can save, edit and run local files, with permission.

Claude Code (Anthropic) | Paid, subscription required

The most capable option for code generation and data analysis. Install: npm install -g @anthropic-ai/claude-code. Requires an Anthropic API key or a paid account. Not free, but the most reliable for complex analytical tasks involving the exported CSV data.

OpenAI Codex CLI (OpenAI) | Paid, or very limited free tier

OpenAI's command-line coding assistant. Install: npm install -g @openai/codex. Requires an OpenAI API key. The free tier is heavily rate-limited; practical use requires a paid account.

Gemini CLI (Google) | Free for personal and research use

Google's CLI assistant. Install: npm install -g @google/gemini-cli. Requires a Google account. Free within generous rate limits. Good general-purpose capability; Claude Code generally outperforms it for code generation.

All CLI tools except Qwen require Node.js v18+. Install Node first at nodejs.org, then use npm install -g to install your chosen tool.

CLI LLMs: Qwen and Practical Workflow

Qwen CLI (Alibaba Cloud) | Free (although this may have changed recently)

A free and capable option from Alibaba Cloud. Generally outperforms Gemini CLI for code generation. Install: pip install qwen-agent or follow the Qwen CLI documentation. Note: developed by a Chinese technology company. Users with data sensitivity concerns should factor this into their choice.

Recommended choice

Try Gemini CLI first: free, no data concerns, good for most tasks. If you need stronger code generation and are comfortable with the cost, Claude Code is the strongest option. Qwen CLI is the most capable free option if you have no reservations about the provider.

Practical workflow with any CLI LLM

1. Export your search results as a CSV. 2. Open a terminal in the same folder. 3. Start your CLI LLM and paste in the CSV filename or drag in the file. 4. Also give it document_metadata_3.5.csv for corpus context. 5. Ask it to write Python to normalise counts, plot trends, run stats, and produce visualisations. 5. You can ask for standalone Python scripts or additional analysis cells in the topic_search notebook.

Important caution

The CLI LLM does not know the NSDDD corpus. You must tell it about unequal document counts, the distinction between sentence counts and document-level prevalence, and any analytical constraints. The LLM writes the code; you make methodological choices.

Use the CLI LLM to write code. Make methodological choices yourself. The LLM executes instructions; it does not supply domain knowledge about the corpus.

Summary

Key Takeaways

1 The system uses semantic embeddings to find meaning across 671 documents.

2 Full sentences produce better results than single words. Think about what meaning you are looking for.

3 Quantitative work requires normalisation: document counts, lengths, and time coverage are unequal.

4 Qualitative work benefits from additional work on exported CSVs to trace terminological variation.

5 AI can write analysis code reliably, yet without care, can ignore vital context.

Neal, A. W. and Gardner, R. B. (2026). NSDDD v3.5. University of Edinburgh. https://doi.org/10.7488/ds/8101